

Computer practical catchment hydrology

Water quantity and quality

Name:

1 Introduction

In the reader you about many hydrological concepts, processes and techniques and you applied some of the basics in the exercises. But when you want to analyze real time series of rainfall or forecast discharge, you need a computer.

In this computer practical you will do some analyses and modelling in the programming environment “R”. For most of you, this will be the first introduction to a programming language, but certainly not the last. Data analysis is a crucial part of most research and a part of all BSc and MSc theses. Learning how to perform these analyses in a programming environment seems like a waste of time at first (the argument “Excel is much easier” is heard very often), but you should look at it as an investment: after a while, it’s much more efficient than mouse-clicking programs such as Excel. Knowing a programming language is also highly valued on your CV.

In the first practical you will learn the basics of R and analyze rainfall. In the second practical you will work on evapotranspiration and rainfall-runoff modelling. Work in couples behind one computer so you can help each other, but fill in the answer sheets separately (so you both practice formulating answers, which is helpful exam training). Upload the filled-in assignments to Brightspace after the practical (this practical is compulsory). After the practical, you will:

- know the basics of R
- have an idea of spatial variability of rainfall and understand the Thiessen polygon method for spatial interpolation of rain gauge data,
- understand how the Budyko curve works and what information you can gain with it and

Before the practical, go through “A (very) short introduction to R”.

Open this file in Adobe Reader and fill in the answers in the blue boxes. **Check now if your answers are saved by entering your name, saving it, closing the file and opening it again.** Remember to save the file often during the practical (CTRL+S). Submit the filled-in version on Brightspace. Each person should fill out and submit his/her own file and run the R scripts, but we recommend to work in groups of 2 (or 3) so you can help each other out. Save the R scripts, so you can use your analyses later in the course and use code for other purposes (e.g. BSc thesis).

2 Precipitation

2.1 Preparation

Read Section 2.5 of the reader about spatial interpolation methods to obtain areal precipitation values from several rain gauges.

Sometimes, the amount of computer code makes it hard to see what the computer is actually computing. Therefore, as a preparation for the computer practical, you will now do an exercise by hand to understand what the computer will do for you later.

Figure 1 shows a map of the headwater (*bovenloop*) of the Issel (*Oude IJssel*) catchment in Germany. In the catchment two discharge measurement stations are operated: in Isselburg and Dämmerwald. Five rain gauges can be used to estimate the catchment average rainfall.

On a certain day, 12 mm of rainfall was measured in Borken, 10 mm in Bocholt, 8 mm in Hamminkeln, 7.5 mm in Raesfeld and 7 mm in Wesel. Draw Thiessen polygons

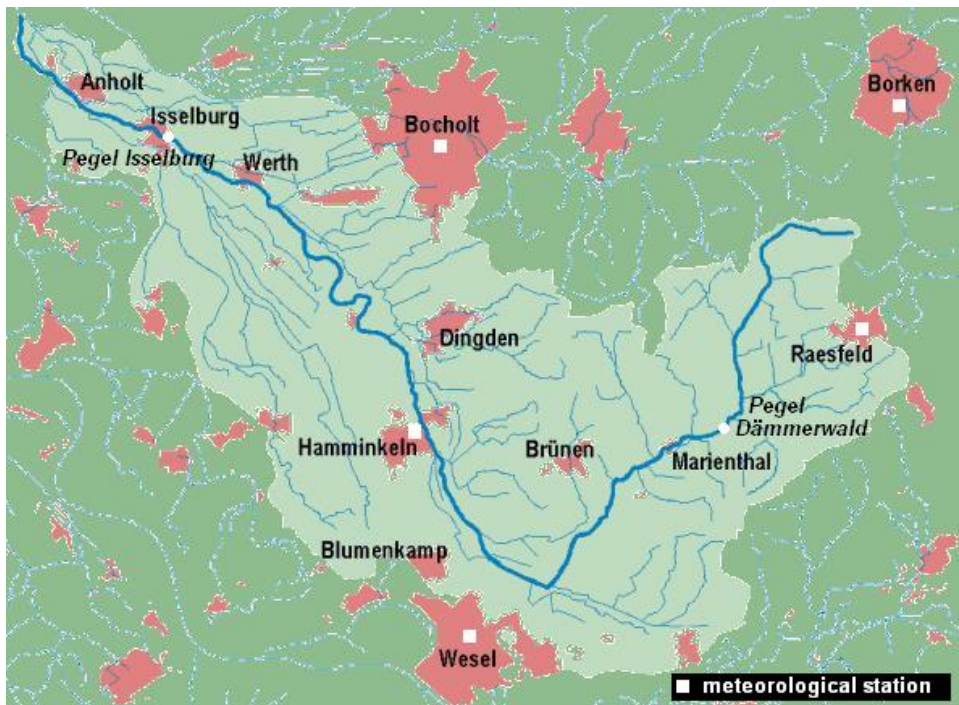


Figure 1: Rain gauges (squares) and discharge stations (circles, *pegel* in German) in the Issel catchment.

and compute the catchment average rainfall of the catchments upstream of Isselburg and Dämmerwald. Estimate the locations of the subcatchment boundaries with the surface water network.



Isselburg: $P = 0.25 \cdot 10$ (Bocholt) + $0.35 \cdot 8$ (Hamminkeln)
+ $0.25 \cdot 7.5$ (Raesfeld) + $0.15 \cdot 7$ (Wesel) = 8.2 mm.
Borken's polygon lies outside both subcatchments (weight 0).

Dämmerwald: lies entirely in the Raesfeld polygon,
so $P = 1 \cdot 7.5 = 7.5$ mm.

1

The answer can be found in Section 2.5.

2.2 Getting started

Open Windows Explorer (Windows; the folder icon on the task bar) or Finder (Mac; smiley face icon), go to **This Computer**, select a logical location on your computer and create a folder called **WQQ** (because the course is called Water Quantity and Quality). Tip: do not make a folder in “My Documents” or in a “Library”, because it is unclear where these folders are located on your computer.

Go to Brightspace. Download and save the zip file called **PCpractical_WQQ.zip** in the **WQQ**-folder you just created. **Unzip** the file, so that it forms a subfolder. You now have: <your logical location>→**WQQ**→**PCpractical_WQQ**.

For the first practical, you need the script **Thiessen.R**. Go to Windows Explorer and double-click this file. If the computer you are working on already knows RStudio, it will open in RStudio automatically. If not, you have to tell your computer that it should open files with extension **.R** in RStudio. To do this, select the program from the list of options it gives you. Maximize the window and increase the size of the script window (top left).

The script follows the same structure as this pdf. You can walk through this pdf – it will tell you when to do something in R.

2.2.1 Programming help

If everything went well, you now have the script open in RStudio. In this script, much is preprogrammed to help you as a new programmer focus on the hydrological content and not on the R code. Sometimes you do have to program some things yourself, but

you only have to do so when there are ... written in the script. In that case, replace the ... with the appropriate commands.

In case of problems, try the following things:

- Look in the environment window (top right) to see what R has in its memory.
- Run the script line by line until you encounter a problem (exception: for-loops and functions have to be run from “for” or the name of the function to the last bracket belonging to that for-loop or function at once).
- Read the text in this assignment sheet.
- Read the comments (green) in the R script. As explained in the R intro, text behind the # is not read by R as a command and is there to explain to you what’s happening.
- Use the R intro. The last pages contain lists of frequently used functions and common errors.
- Look in the R help function.

2.3 Thiessen polygon maps

2.3.1 Getting started

Go to the R script `Thiessen.R`. Under the header “Getting started”, change the working directory to where you saved all files for this practical:

“`your location\WQQ\PCpractical_WQQ\`”. You can also change the working directory through the menu (Session $\rightarrow$ Change working directory $\rightarrow$ To source file location). Run the lines in the section “Getting started” by putting your cursor in the script window and pressing CTRL+ENTER. You have to install some packages necessary for the graphical analyses (see comments in the script).

2.3.2 Read data

Run the “Read data”-part of the script. In this part, data are read from some files, which are all stored in the data subfolder. Try to understand roughly what R is doing by reading the comments (*commentaar*) and looking at the commands (*commando's* – don’t try to understand all of them). Also look what happens in the environment window (top right) when you read the data.

► Go to Windows Explorer (or a similar program) and open the file `XYZ_Raingauges_Ourthe_2002.txt` in a simple text editor like Notepad. Guess what the columns mean.

ID:

2 Name (code) of the rain gauge, e.g. GEMM, WARE

X:

3 x-coordinate (easting) of the gauge, UTM zone 31N [m]

Y:

4 y-coordinate (northing) of the gauge, UTM zone 31N [m]

Z:

5 Elevation of the gauge [m above sea level]

► Open `P_Hour_Ourthe.txt` in Notepad. What is written in the different columns (give unit if possible)?

First column:

6 Date and hour, format `yyyymmddhh`

Other columns:

7 Hourly rainfall sum at each of the 42 rain gauges [mm/h]

► Open `catchment_1km.asc` in a text editor like Notepad. (You may have to tell your computer that you want files with extension `.asc` to open in Notepad.) This file can be seen as a map with all pixels in and around the catchment. What do 0 and 1 mean?

0:

8 Pixel (1 km x 1 km) outside the catchment (NODATA value)

1:

9 Pixel inside the catchment

The files starting with `angleurBoundary` are called shapefiles and together they define the boundary of the catchment. You don't have to look at them, because they are quite complex and don't open in text editors.

2.3.3 Compute polygons

In this section, you will compute the rainfall per time step per pixel with Thiessen polygons. To compute this, you need a function. This function is pre-programmed and

stored in another R script: `Thiessen_function.R`. Read Section 11.3 of the “Very short introduction to R” about functions and reread Section 5 about scripts.

The function computes Thiessen polygons, makes a rainfall map for a certain day and hour and gives the catchment average precipitation sum.

► Open this script in RStudio, read the comments (green), but don't pay too much attention to the commands (black; quite complicated for starting programmers).

The nice thing about functions, is that you can call them with different arguments (in this case `date_select`) to do the same computations with different numbers very easily. The script gives some hints about how to call the function. **Do not change anything in the function itself.**

You can call the function by typing `thiessen(2002122101)` to make a map of the rainfall sum between 0 and 1 o'clock on 21 December 2002. When you call the function like this, R will virtually fill in 2002122101 everywhere where `date_select` is written in the function.

Try several hours. For example, follow the cloud that passed between 10 Nov. 19⁰⁰ and 11 Nov. 04⁰⁰. Look at the maps you created. Click on the arrows in the plot window to see previous plots.

► What do the plusses mean in the figure?

10 The locations of the rain gauges

► And the lines?

11 Thin lines: Thiessen polygon boundaries; thick line: catchment boundary

2.3.4 Determine fractions

Making a map for each time step to compute the catchment average precipitation is computationally unnecessarily heavy. Because the locations of the polygons do not change in time, you can determine the contribution (weight) of each polygon once and use those contributions to compute the catchment average precipitation for the rest of the time series.

Read the comments in this part of the script and run it. You created a vector called `weights` and put this in a textfile called `weights_Thiessen.dat`.

► Open the text file `weights_Thiessen.dat` in a text editor. Which station has the largest contribution and how large is it?

12 VIEL, with a contribution of 0.093 (9.3%)

2.4 Time series

In this section you will construct a long time series of catchment average precipitation and analyse them. For these analyses you have to program some things yourself. Replace the “...” in the script with the appropriate commands (or add lines with commands) and run them with CTRL+ENTER.

2.4.1 Differences between stations

► Name three important factors which cause spatial variability of rainfall.

1.

13 Elevation / orography (lifting of air, cooling, condensation)

2.

14 Distance to the sea and wind direction (windward vs. lee side)

3.

15 Type of rainfall: local convective showers vs. widespread frontal rain

Compute yearly rainfall sums for all 42 stations and make barplots.

► Compute the mean and standard deviation of all yearly rainfall sums. Also compute the maximum and minimum sums.

Mean:

16 1216 mm

St. dev.:

17 207 mm

Max:

18 1648 mm (BOUI)

Min:

19 872 mm (ROCH)

- What do these numbers and the barplot say about the spatial variability of rainfall in the Ourthe catchment?

Large spatial variability: the wettest gauge gets almost twice as much as the driest (CV about 17%), so one gauge is not representative.

20

- Are these sums large or small compared to yearly rainfall sums in the Netherlands?

Large: the Netherlands gets about 800-900 mm/y.

21

Plot the annual rainfall sums as a function of the elevation of the gauges.

- What relation can you see between the two variables? Can you explain this relation physically?

Rainfall increases with elevation ($r = 0.54$, about $+0.8$ mm/y per m). Orographic effect: air forced up over the Ardennes expands and cools, water vapour condenses, so more rain falls at higher elevations.

22

2.4.2 Compute catchment average

In this part of the script the catchment average precipitation is computed in 3 ways. Read the comments and commands and try to understand what R is doing. To compute the catchment average precipitation with the Thiessen polygon method, R uses the vector `weights` you created in section 2.3.4.

- What do `P_th` and `P_ar` mean?

`P_th`:

Catchment average rainfall per hour with the Thiessen polygon method

23

`P_ar`:

Catchment average rainfall per hour with the arithmetic mean of all gauges

24

- Explain the difference between the two techniques.

The arithmetic mean gives all gauges the same weight. Thiessen weights each gauge by the area of its polygon inside the catchment, so gauges representing large areas count more and gauges outside the catchment count less or not at all.

25

Many catchments are poorly instrumented. If there is only one rain gauge, you have to assume its measurements to represent the whole catchment. To investigate the effect of such data shortage, choose (arbitrarily) one of the rain gauges and treat it as the third method to compute catchment average precipitation: P_{one} . Program the commands to select the precipitation time series measured at this station. Read Section 6.3 of the R intro on how to select one column from a data frame.

► Which station did you choose?

26 VIEL (the gauge with the largest Thiessen weight)

2.4.3 Differences between methods

In this section you will analyse the differences between the time series of the three catchment average methods P_{th} , P_{ar} and P_{one} .

► What are the yearly sums and maximum hourly rainfall sums of P_{th} , P_{ar} and P_{one} ?

	P_{th}	P_{ar}	P_{one}
Yearly sum	27 1312	28 1216	29 1278
Max. hourly sum	30 5.2	31 4.7	32 16.5

► How can you explain the differences?

33
 P_{ar} is lower: many gauges in the lower, drier areas (or outside the catchment) get the same weight as gauges that represent large areas. One gauge catches the full intensity of a local shower, while averaging over many gauges smooths out peaks.

Plot P_{th} as a function of P_{ar} . This figure illustrates how the daily sums of the two methods compare.

► Describe the differences in rainfall from the two methods.

34
 Points are close to a straight line ($r = 0.92$) but with slope 1.08: P_{th} is about 8% higher than P_{ar} , with scatter for individual (shower) hours.

► Do you think that it paid off to use Thiessen polygons instead of the arithmetic mean method for the Ourthe catchment? Why?

35
 Yes, the yearly sums differ by about 100 mm (8%). Gauges are unevenly spread and some lie outside the catchment; Thiessen corrects for that.

► Of course this difference is different for different catchments. What determines the difference between the Thiessen and arithmetic mean method?

How evenly the gauges are spread over the catchment (clustering, gauges outside it) and the spatial variability of the rainfall itself (e.g. elevation differences).

36

Plot P_{th} as function of P_{one} to check if the catchment average precipitation can be predicted by one rain gauge.

► Does the rain gauge you chose give a good estimate of the catchment average? Why?

No. The yearly sum is close, but per hour the scatter is large ($r = 0.77$): peaks are much too high or missed, and in 64% of rainy hours VIEL measured nothing. One point cannot represent local showers.

37

2.4.4 Consequences of rainfall estimation

This last question is a thought-experiment. Suppose you use data from the single gauge (P_{one}) instead of the correct catchment average rainfall as input in a rainfall-runoff model to simulate discharge.

► Will the wrong rainfall input lead to an underestimation, overestimation or correct estimation of discharge? Explain how the rainfall error propagates (*zich voortplant*; “how one error leads to an error in the next step”)?

VIEL underestimates the yearly sum, so discharge is underestimated. The error propagates: too little water goes into soil moisture and groundwater storage. ET hardly changes, so almost the whole rainfall error ends up in the discharge (a larger relative error in Q than in P). Peaks are also wrong in timing and height.

38

2.5 Answer preparation exercise

See the sketch in Figure 2 to estimate the percentage of each subcatchment belonging to each rain gauge. Note that it is difficult to determine the catchment boundaries based on the surface water network in areas where water courses are connected.

Estimate the contribution of each rain gauge to the catchment average precipitation of both subcatchments. You can do it roughly by eye or more accurately by dividing the figure into squares and counting the squares belonging to each rain gauge (that is in fact what you will do in the computer practical).

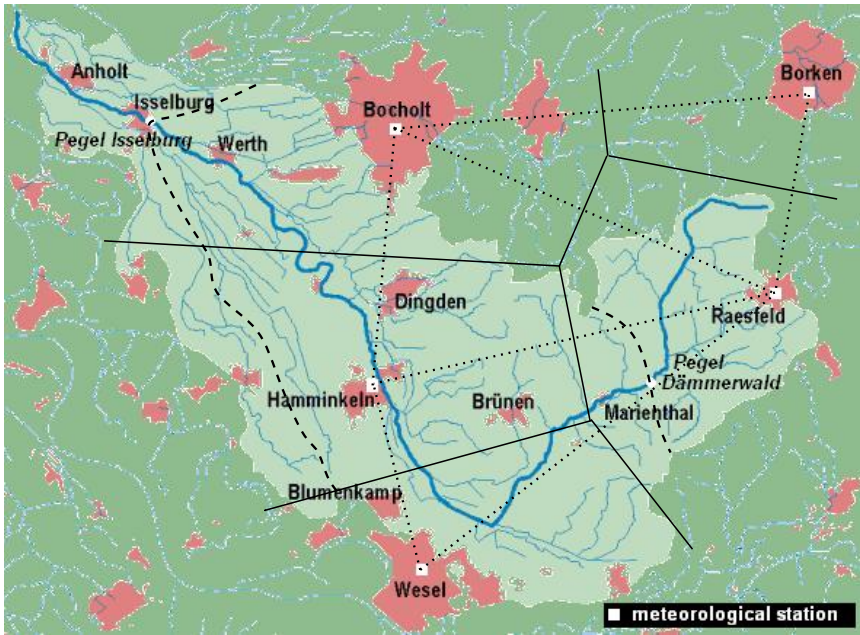


Figure 2: Thiessen polygon in the Issel catchment. Dashed lines: catchment boundaries. Dotted lines: lines connecting rain gauges. Solid lines: perpendicular bisectors through the dotted lines, which separate areas belonging to each rain gauge.

	Isselburg	Dämmerwald
Bocholt	25	0
Borken	0	0
Hamminkeln	35	0
Raesfeld	25	100
Wesel	15	0

$$P_{\text{Isselburg}} = 0.25 \cdot P_{\text{Bocholt}} + 0.35 \cdot P_{\text{Hamminkeln}} + 0.25 \cdot P_{\text{Raesfeld}} + 0.15 \cdot P_{\text{Wesel}}$$

$$P_{\text{Isselburg}} = 0.25 \times 10 + 0.35 \times 8 + 0.25 \times 7.5 + 0.15 \times 7 = 8.2 \text{ mm}$$

$$P_{\text{Daemmerwald}} = 1 \cdot P_{\text{Raesfeld}} = 7.5 \text{ mm}$$

3 Evapotranspiration

3.1 Preparation

Read section 3.5 of the reader about the Budyko curve.

In the PC practical you will get daily precipitation, potential evapotranspiration and discharge data from several catchments around the world (see Table 1). Guess for each catchment what the yearly precipitation and actual evapotranspiration sums are (in mm) and write the numbers in Table 1. Also indicate whether you expect that the evapotranspiration reduction (difference between potential and actual evapotranspiration) is large or small.

During the computer practical you will use data to get the answers and fill in Table 2.

Table 1: Guesses of water balance terms and evapotranspiration reduction (difference between ET_{act} and ET_{pot}).

Catchment		P [mm y ⁻¹]		ET_{act} [mm y ⁻¹]		ET reduction (large or small)	
Guadiana (Spain)	39	500	40	400	41	large	
Hupsel Brook (Netherlands)	42	800	43	500	44	small	
Mahakam (Indonesia)	45	3000	46	1200	47	small	
Metuje (Czech rep.)	48	700	49	450	50	small	
Narsjø (Norway)	51	800	52	300	53	small	
Ourthe (Belgium)	54	1000	55	500	56	small	
Plynlimon (Wales, UK)	57	2500	58	450	59	small	
Rietholzbach (Switzerland)	60	1500	61	550	62	small	

3.2 Getting started

Close the script `Thiessen.R` and open `Budyko.R` in RStudio. Set the working directory to the folder where you stored all files and read the data.

► Open one of the files called `PEQ_..._day.dat` in a text editor (all these files have more or less the same format). Guess what the columns mean and their units.

P:

63 Precipitation [mm/d]

ET_{pot}:

64 Potential evapotranspiration [mm/d]

Q:

65 Discharge as water depth over the catchment [mm/d] (T = temperature [°C])

3.3 Water balance terms

Compute (with R) yearly sums of P , ET_{pot} and Q and compute ET_{act} by closing the water balance. Note that for the Narsjø catchment, the only available rain gauges are in valleys outside the catchment, while the elevation reaches 1600 m above sea level (remember the practical on precipitation).

► Which water balance term did you neglect when you assumed that $P = Q + ET_{\text{act}}$?

66 The storage change (dS/dt) in soil, groundwater, surface water and snow

► Why is it valid to assume that this term is negligible in this case?

67 The data cover many years (3 to 40). Over such long periods the storage at the end is about the same as at the start, so dS is negligible compared with P, ET and Q.

► In which cases is this assumption not valid?

68 For short periods (days, months, one year), with a trend in storage (groundwater extraction, melting glaciers) or with large groundwater flows across the catchment boundary.

Write the yearly sums of P , Q , ET_{pot} and ET_{act} in Table 2 (in whole mm) and compare it to your guesses. Hint: the R code makes a table called `WB`, so you can just open it and copy the numbers.

Table 2: Water balance terms and evapotranspiration limitation factors found from data (to be filled in during the practical).

Catchment	P [mm y ⁻¹]	ET_{act} [mm y ⁻¹]	ET reduction (large or small)	ET limiting factor
Guadiana (Spain)	69 452	70 433	71 large	72 water
Hupsel Brook (Netherlands)	73 805	74 477	75 small	76 energy
Mahakam (Indonesia)	77 3256	78 453	79 large	80 energy
Metuje (Czech rep.)	81 750	82 385	83 medium	84 energy
Narsjø (Norway)	85 593	86 227	87 unreliable	88 energy
Ourthe (Belgium)	89 995	90 543	91 small	92 energy
Plynlimon (Wales, UK)	93 2452	94 460	95 small	96 energy
Rietholzbach (Switzerland)	97 1458	98 437	99 small	100 energy

► Which values did you over- or underestimate?

101 Narsjø: P too low (valley gauges, snow), so $ET_{act} < 0$. Mahakam: ET_{act} too low.

► How large is the Dutch precipitation sum compared to the other catchments?

102 Among the lowest (805 mm/y); only Guadiana and Narsjø are lower.

► Do you promise never to complain about the Dutch weather again?

103 Yes! It doesn't rain that much in the Netherlands, it just rains often.

3.4 Budyko curve

Write the yearly sums in four vectors. Find out what is plotted on the x - and y -axes of a Budyko curve and program this in the script. Lines for $ET_{act} = ET_{pot}$ and $ET_{act} = P$

are preprogrammed. Write in Table 2 whether the catchments are water or energy limited.

- Of which catchments did you guess the limiting factor wrongly? What did you under- or overestimate?

104 Metuje (energy limited, not water limited) and Mahakam (energy limited but large ET reduction). ETact in wet, cool catchments was overestimated.

- Which catchment is the most energy limited?

105 Plynlimon ($ET_{pot}/P = 0.22$)

- And the most water limited?

106 Guadiana ($ET_{pot}/P = 2.78$, $ET_{act}/P = 0.96$)

Add the curve of Zhang (Eq. 1) to the plot. Some help is provided in the script. Use for w 1.70, 2.63 and 5.00 to obtain Figure 7.30 in the reader.

$$\frac{ET_{act}}{P} = 1 + \frac{ET_{pot}}{P} - \left[1 + \left(\frac{ET_{pot}}{P} \right)^w \right]^{1/w} \quad (1)$$

- How do the lines describe the points?

107 Most points lie close to the curves (around $w = 2.63$); Mahakam and Narsje don't fit due to data errors.

- **BONUS** In the Metuje catchment about 50% of the land is covered with forest. What would happen to the Metuje-point in the Budyko-curve when the forest would be removed and replaced with pastures?

108 ETact decreases (shallower roots, less interception), so the point moves down to a curve with a lower w and Q increases.

- **BONUS** How can you explain this change physically in terms of soil moisture stress and evapotranspiration reduction?

109 Grass roots reach less soil water, so soil moisture stress and ET reduction occur earlier in dry periods; interception evaporation is also lower.

3.4.1 Seasonality

Compute monthly means of P , ET_{pot} and Q for all catchments and make a barplot for each catchment.

- Why can't you use ET_{act} instead of ET_{pot} here?

ETact = $P - Q$ only holds for long periods. Per month the storage change is large (storage in winter, depletion in summer), so $P - Q$ is not ETact.

110

- What can you see in the discharge record of the Narsjø?

Very low discharge in winter and a large peak in May-June (8 mm/d, higher than P): precipitation is stored as snow and released as snowmelt.

111

- The potential evapotranspiration in the Mahakam catchment is the same for every month. This is an approximation, but this approximation is reasonable because evapotranspiration hardly shows seasonality in Indonesia. Why?

Near the equator radiation, day length, temperature and humidity are almost constant over the year, so the energy available for ET is constant.

112

- In all catchments $ET_{\text{pot}} + Q > P$ during part of the year. Explain what the consequences of such periods are for storage and actual evapotranspiration.

More water leaves than enters, so storage (soil moisture, groundwater) decreases. The soil dries out, causing water stress, so $ET_{\text{act}} < ET_{\text{pot}}$.

113

- Explain what the consequences of such periods are for the limiting factor (energy/water).

During these periods even an energy-limited catchment becomes (temporarily) water limited, e.g. in summer.

114

4 Runoff

4.1 Preparation

Read sections 8.4.3 and 9.5 in the reader about baseflow separation and the unit hydrograph.

Assume the unit hydrograph for a certain catchment for 1 mm of rainfall and hourly time step has the following ordinates: 0.3, 0.6 and 0.1 mm (Fig. 3, left panel). On this catchment it rained for four hours, with sums of 10, 2, 1 and 8 mm (Fig. 3, right panel).

► Compute the discharge response to this rainfall event (assume that all rainfall will be discharged and that there is no discharge from previous events). In other words, compute the convolution hydrograph for this particular event.

$$Q_1 =$$

$$Q_1 = 10 \cdot 0.3 = 3.0 \text{ mm/h}$$

$$Q_2 =$$

$$Q_2 = 2 \cdot 0.3 + 10 \cdot 0.6 = 6.6 \text{ mm/h}$$

$$Q_3 =$$

$$Q_3 = 1 \cdot 0.3 + 2 \cdot 0.6 + 10 \cdot 0.1 = 2.5 \text{ mm/h}$$

$$Q_4 =$$

$$Q_4 = 8 \cdot 0.3 + 1 \cdot 0.6 + 2 \cdot 0.1 = 3.2 \text{ mm/h}$$

$$Q_5 =$$

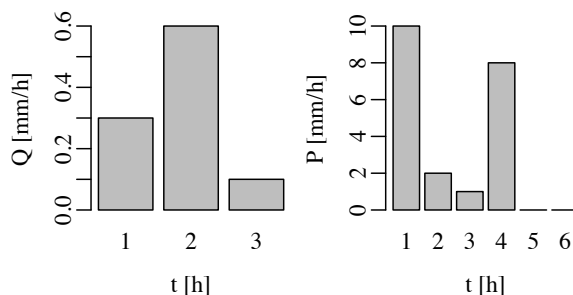


Figure 3: The 1-hour 1-mm unit hydrograph (left) and rainfall event (right) from the exercise.

119 $Q_5 = 8 \cdot 0.6 + 1 \cdot 0.1 = 4.9 \text{ mm/h}$

$Q_6 =$

120 $Q_6 = 8 \cdot 0.1 = 0.8 \text{ mm/h}$

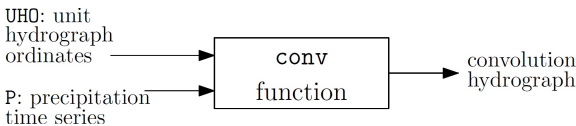
The answer can be found in Section 4.7.

4.2 Using the unit hydrograph

Close the script `Budyko.R` and open `unit_hydrograph.R` in RStudio.

► Define the unit hydrograph ordinates from the preparation exercises as the vector `UHO_example` in the script and make a barplot of `UHO_example`. Do the same with the rainfall series (call it `P_example`).

Instead of adding up all elements by hand as you did in the exercise, you can also use a function. Reread Sec. 11.3 in the R intro about functions. In the script the function `conv` is programmed to compute the convolution hydrograph. Schematically, the function works like this:



The nice thing about this function is that you can investigate what happens to the convolution hydrograph when you use different input very easily: you can simply call the function with different arguments (the vectors `P` and `UHO`) and compare the results (do not change anything in the script again).

- Try to understand what happens inside this function. For more explanation on for-loops and if-statements, see the R introduction.
- Run the function `conv` with `UHO_example` and `Q_example`. Does it yield the same result as your manual computations?

121 Yes: 3.0, 6.6, 2.5, 3.2, 4.9, 0.8 mm/h (sum 21 mm = rainfall sum)

4.3 Fitting a unit hydrograph

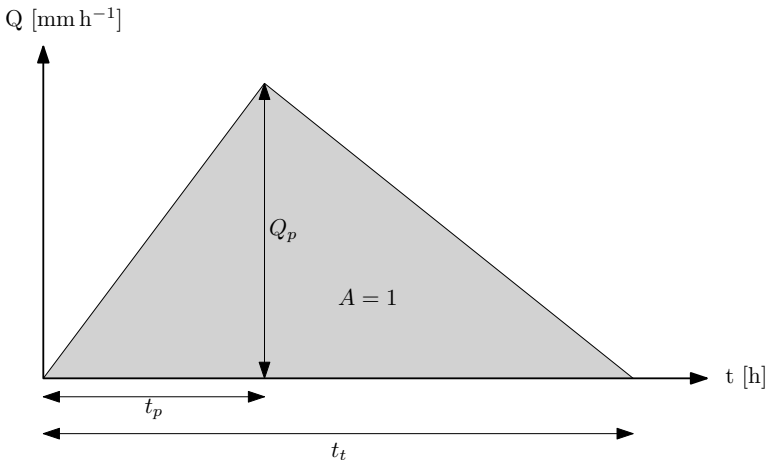
For most catchments you don't know the unit hydrograph on forehand. There are two ways to obtain a unit hydrograph:

- From data (bottom-up approach)
- From theory (top-down approach)

You will apply both methods to obtain unit hydrographs for the Hupsel Brook catchment.

4.4 Bottom-up: the triangular UH

The unit hydrograph can have any shape. In the true bottom-up approach, the shape is one of the things to find out by fitting the model to the observations. A more complex shape requires more parameters to describe it. To restrict the number of parameters you have to fit today, you will use a triangle-shaped unit hydrograph, which can be defined by only 2 parameters. In the Figure below, the unit hydrograph is plotted with time (t) on the x -axis and the discharge (Q) on the y -axis.



You can see that:

- The whole discharge event lasts for t_t hours: the total time (the base of the triangle).
- The peak occurs at t_p hours after the rain event: the time to peak. The corresponding discharge is Q_p : the height of the triangle.
- The area under the triangle must be unity (1), because all effective rainfall must be discharged.

To compute the peak discharge Q_p , we need to know that the area of a triangle is $A = \frac{1}{2}BH$. In this case $A = 1$, $B = t_t$ and $H = Q_p$, so $1 = \frac{1}{2} \times t_t \times Q_p$, leading to

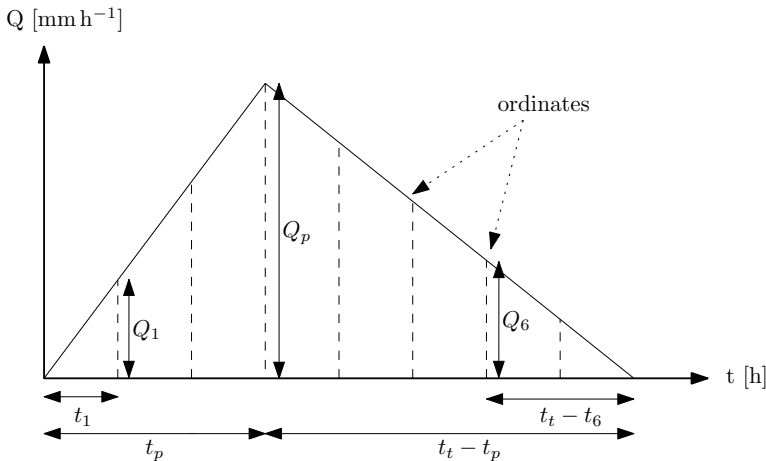
$$Q_p = \frac{2}{t_t}.$$

Unit hydrograph ordinates (the discharge Q_t at certain time t since the unit rainfall) can be computed from ratios within the triangle (see Figure below). Use the left angle before the peak:

$$\frac{Q_t}{t} = \frac{Q_p}{t_p}.$$

Use the right angle after the peak:

$$\frac{Q_t}{t_t - t} = \frac{Q_p}{t_t - t_p}.$$



► Express the discharge Q_t in t , t_t , t_p and Q_p :

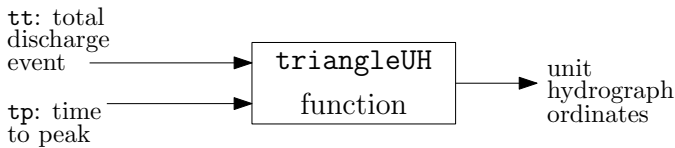
if $t < t_p$, then $Q_t =$

$Q_t = Q_p * t / t_p$

if $t \geq t_p$, then $Q_t =$

$Q_t = Q_p * (t_t - t) / (t_t - t_p)$

In the R script you will find a function called `triangleUH` (partly preprogrammed), which computes unit hydrograph ordinates from t_t and t_p :



► Finish this function by replacing the ... with appropriate commands. Test the function by calling it (don't forget to run the whole function, including its name and the curly brackets) with random values of t_t and t_p and making a barplot of the ordinates.

4.4.1 Calibration

When you calibrate a model, you use data from a certain period and adjust the parameters so that the model run fits the observations (usually discharge) best. This is also called parameter estimation or optimization. With this procedure you hope to find parameter values for a certain catchment, which you can use for all future simulations.

Run the lines under Read data to read observed time series of rainfall and discharge (both in mm h^{-1}). Select precipitation and discharge from one event on which you will fit your model. You can cut one event from the entire data series by selecting only those elements of the vectors P and Q for which the date is between the start and the end dates (see the script). Examples of events are given in the script, but you can also find events yourself.

► Which period did you use?

124 4 Feb 2002 - 9 Feb 2002 (start = 2002020400, end = 2002020900)

4.4.2 Baseflow separation

First separate the baseflow from the direct runoff.

► Why is this necessary?

125 The UH only describes the direct (fast) response to rainfall. Baseflow comes from groundwater that was already there before the event, so it must be removed.

As a quick estimate, assume that the baseflow is constant, at the minimum discharge during the event. Use the script to separate direct runoff from baseflow.

► How large is the precipitation sum for your event?

126 34.5 mm

► And the direct runoff sum?

127 13.3 mm (baseflow 0.023 mm/h)

► What is $\sum DR / \sum P$ (a kind of runoff ratio)?

128 0.39

► What does this runoff ratio tell you about the initial conditions (the wetness of the catchment before the rainfall event)?

129 Only 39% became direct runoff; 61% was stored. The catchment was not fully wet: the soil still had storage capacity.

4.4.3 Effective precipitation

As you see, the runoff ratio (RR) is not one, so the assumption that all rainfall turned into direct runoff is not completely valid.

► Will the discharge be over- or underestimated in the model due to this error? Why?

130 Overestimated. The UH assumes that all rain becomes direct runoff (ordinates sum to 1), but only 39% does, so modelled discharge would be about 2.6 times too large.

To avoid this error, correct for effective precipitation. Assume that a fixed fraction of the rain will be absorbed by the soil. This fraction is $1 - RR$, so you can compute the time series of effective precipitation from the time series of precipitation as $P_{\text{eff}} = P \times RR$.

► What is the difference between precipitation and effective precipitation?

131 Precipitation is all the rain that falls. Effective precipitation is only the part that becomes direct runoff; the rest (infiltration, storage, interception, evaporation) is a loss for the event.

► Why should you use P_{eff} for the unit hydrograph approach?

132 The UH only distributes the input over time and conserves volume. Only water that actually becomes direct runoff may be used as input, otherwise the volume of the modelled discharge is wrong.

- When is the difference between P and P_{eff} large? Why?

When the catchment is dry (summer, low groundwater), on permeable sandy soils or with low rain intensity: much rain can infiltrate and be stored. In wet conditions or with intense rain the difference is small.

133

4.4.4 Optimization

You can use the ordinates from the `triangleUH` function and rainfall data as input in the `conv` function to obtain a convolution hydrograph. This convolution hydrograph is the modelled direct runoff. Plot measured and modelled direct runoff in one graph to see how well the model simulates the observed discharge.

Run the `triangleUH` and `conv` functions with different values of t_t and t_p . Compare (by eye) the modelled and observed direct runoff.

- Which values of t_t and t_p lead to the best results?

$t_t = 87$ h, $t_p = 11$ h

134

- What were your selection criteria?

Highest Nash-Sutcliffe efficiency, and by eye: timing and height of the peak, shape of the recession and volume.

135

- Imagine that you are working for a water authority that wants the best model to simulate the height of the discharge peak, which will be used to design flood protection structures. Would you advise the parameter values above or do you need different parameters for this application?

No. The peak is underestimated (0.25 vs 0.29 mm/h) because NS weighs the whole event. Calibrate on peak height, using several large events.

136

- Could you use those parameters also if your main aim is to predict the timing of the peak well (e.g. for evacuation)?

No, the modelled peak is about 5 h too early. Calibrate t_p on peak timing.

137

4.4.5 Goodness of fit

You compared the observed and modelled time series by eye or use quantitative methods to evaluate the goodness of fit. Examples of goodness of fit measures are the Sum of Squares and the Nash-Sutcliffe Efficiency:

$$NS = 1 - \frac{\sum (Q_{\text{obs}} - Q_{\text{mod}})^2}{\sum (Q_{\text{obs}} - \bar{Q}_{\text{obs}})^2}$$

When $NS = 1$, the fit is perfect, when $NS = 0$ an average discharge value would be as good as the model and when $NS < 0$ the model is really bad. The Nash-Sutcliffe efficiency is pre-programmed.

- What is the Nash-Sutcliffe efficiency for your best fit?

138 `NS = 0.93`

- Do you think it is high? Why?

139 `Yes, it is high (1 = perfect, > 0.7 is good). But it is the calibration event, to which the parameters were fitted, so a high value is expected.`

4.5 Validation

The unit hydrograph of a certain catchment is supposed to be the same for all rainfall events. To investigate whether the parameters you fitted are case-specific or generic, you should run the model with the same parameters for a different rainfall event. This is called validation.

Select a validation period (under Select data in the script again), separate the baseflow and run the `conv` function.

This seems a repetition of what you did before, but it is fundamentally different, because you are not building a model, but checking it. Validation is an important part of any model study.

- Which period did you use?

140 `29 Dec 2002 - 2 Jan 2003 (start = 2002122900, end = 2003010200)`

- Was the model able to simulate the discharge? What went well and what not?

141 `Only partly. The volume is correct, but the peak is much too low (0.30 vs 0.93 mm/h) and about 7 h too late.`

- Why do you think the model was not able to simulate the things that went wrong?

The UH assumes a linear, time-invariant response, but the response depends on wetness and rain intensity (more fast flow in this wet period). The constant loss fraction and baseflow are also simplifications.

142

- What was the Nash-Sutcliffe efficiency? What can you say about it?

NS = 0.29, much lower than in calibration: the parameters are event-specific.

143

4.6 Top-down approach: the j -model

Most catchments around the world are ungauged basins, without discharge data to fit the model to. In those cases you can derive a unit hydrograph from theory and catchment characteristics, such as soil properties.

In this exercise you will use the j -model. In Chapter 4 of the reader, the reservoir coefficient j is introduced. In Section 9.7.3 the Wageningen model is explained, in which the j -model plays an important role.

The j -model was derived in the 1950s by a former professor of the Hydrology Group in Wageningen, Prof. Kraijenhoff van de Leur. The concept is that if the catchment behaves as an infinitely long strip of flat land with ditches on both sides and with drainage length (the distance from the middle of the field to a ditch) L [m], porosity μ [-], hydraulic conductivity k [m/d] and aquifer depth D [m], a reservoir coefficient j [d] can be found from:

$$j = \frac{1}{\pi^2} \frac{\mu L^2}{k D} .$$

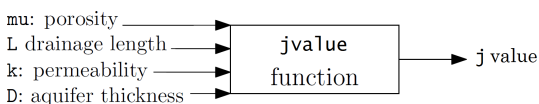
The unit hydrograph of this model is:

$$u(0, t) = \frac{8}{\pi^2} \cdot \frac{1}{j} \sum_{n=1,3,5}^{\infty} e^{-\frac{n^2 t}{j}} .$$

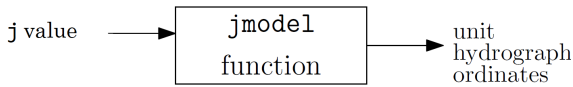
If you only use the first term of the Fourier series, this becomes:

$$u(0, t) = \frac{8}{\pi^2} \cdot \frac{1}{j} e^{-\frac{t}{j}} .$$

There are two functions partly preprogrammed. The function `jvalue` computes the reservoir coefficient from the four catchment characteristics:



The function `jmodel` computes unit hydrograph ordinates from the reservoir coefficient:



► Finish these functions in the script.

4.6.1 Parameter estimation

The catchment area of the Hupsel Brook catchment is 6.5 km^2 . The brook and its tributaries are in total 15 km long. The value of μ ranges from about 0.05 to 0.1, k from 1 to 5 m d^{-1} , and D from 1 to 8 m.

► Estimate the drainage length L [m] from the catchment area A [m^2] and the total channel length ll [m]. To do this, you can approximate the catchment as a rectangle with area A and a river in the middle with length ll .

144 $A = 2 * L * ll$, so $L = A / (2 * ll) = 6.5e6 / (2 * 15000) = 217 \text{ m}$

Use the `jvalue`-function to compute the largest and smallest possible values of j based on the parameters given above.

► What are the smallest and largest value of j ?

145 Smallest: 6.0 d = 143 h ($\mu=0.05$, $k=5$, $D=8$); largest: 477 d = 11451 h ($\mu=0.1$, $k=1$, $D=1$)

Run the `jmodel` function with a j value in this range.

► Plot the unit hydrograph ordinates and compare them to the ones from the triangular UH. What are the differences?

146 `j-model` ($j = 716 \text{ h}$): no rising limb, highest ordinate at $t = 1$, then a slow exponential decay over hundreds of hours. The triangle peaks at 11 h and ends at 87 h.

4.6.2 Validation

Run the model with data from the same event you used before and your choice of j .

► Was the model able to simulate the discharge?

No. With $j = 716$ h, $NS = -0.96$: the modelled direct runoff is far too low and spread out over too long a period, with no peak.

147

► The j -model is based on an emptying groundwater reservoir and actually represents the baseflow. Which rainfall-runoff processes (flowroutes) are not included in the model and make up the direct runoff?

Fast flow routes: overland flow (infiltration excess and saturation excess), direct rain on the channels, interflow (shallow subsurface flow) and flow through tile drains.

148

4.6.3 Calibration

Because in this case you do have discharge data from the catchment, you can estimate the parameters of the j -model by optimization. Try different values for j .

► Which value did you find?

$j = 40$ h

149

► What was the Nash-Sutcliffe efficiency? Was it higher than for the triangular unit hydrograph?

$NS = 0.73$, lower than the triangular UH (0.93)

150

► The timing of the peak is not well predicted by the j -model. Why?

The j -model UH has its maximum at $t = 1$ and no rising limb (no travel time), so modelled discharge responds immediately and peaks too early; a fitted j must also be small to get the peak height.

151

4.7 Answer preparation exercise

$$Q_1 = P_1 \times UHO_1$$

$$Q_2 = P_2 \times UHO_1 + P_1 \times UHO_2$$

$$Q_3 = P_3 \times UHO_1 + P_2 \times UHO_2 + P_1 \times UHO_3$$

$$Q_4 = P_4 \times UHO_1 + P_3 \times UHO_2 + P_2 \times UHO_3$$

$$Q_5 = P_4 \times UHO_2 + P_3 \times UHO_3$$

$$Q_6 = P_4 \times UHO_3$$

$$Q_1 = 10 \times 0.3 = 3.0 \text{ mm}$$

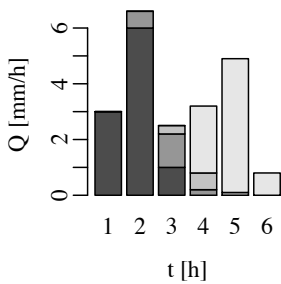
$$Q_2 = 2 \times 0.3 + 10 \times 0.6 = 6.6 \text{ mm}$$

$$Q_3 = 1 \times 0.3 + 2 \times 0.6 + 10 \times 0.1 = 2.5 \text{ mm}$$

$$Q_4 = 8 \times 0.3 + 1 \times 0.6 + 2 \times 0.1 = 3.2 \text{ mm}$$

$$Q_5 = 8 \times 0.6 + 1 \times 0.1 = 4.9 \text{ mm}$$

$$Q_6 = 8 \times 0.1 = 0.8 \text{ mm}$$



As you noticed, this is time-consuming and boring work, especially when you work with long time series, complex unit hydrographs or small time steps. Therefore you are happy to use your good friend R again to compute the same thing faster.